
User-Generated Content and Immersive Brand Experience: A Text Mining Approach

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Abstract: With the rapid development of the Metaverse and virtual reality (VR) technologies, immersive brand experience has become an increasingly important mode of interaction between brands and consumers. Unlike traditional questionnaire-based approaches, which rely on structured and often static data, user-generated content (UGC) provides more authentic, dynamic, and context-rich insights into consumers' real experiences and emotional responses in actual usage scenarios. Against this backdrop, this study focuses on user reviews of three mainstream VR devices—Meta Quest, HTC Vive, and PICO—collected from Amazon and JD.com, and applies natural language processing (NLP) techniques to conduct a systematic and data-driven analysis of textual information.

Specifically, this study employs multiple text mining techniques, including Chinese word segmentation, word frequency analysis, TF-IDF keyword extraction, sentiment analysis, and word cloud visualization, to mine large-scale user reviews and extract meaningful patterns from unstructured data. Through this analytical framework, the study identifies key dimensions of immersive brand experience, including immersion, interactivity, device performance, and user comfort, as well as the overall emotional tendencies expressed by users in real-world usage contexts.

The findings indicate that user evaluations are predominantly positive, reflecting a generally high level of satisfaction with VR devices and immersive experiences. At the same time, negative feedback reveals functional concerns related to product performance, usability, and physical comfort, suggesting that technological limitations still constrain user experience. Furthermore, comparative analysis across brands demonstrates that different technological characteristics lead to distinct patterns of user perception and evaluation, highlighting the heterogeneous nature of immersive brand experience in virtual environments.

By adopting a UGC-based analytical approach, this study uncovers the semantic structure and emotional mechanisms underlying immersive brand experiences, thereby extending existing research beyond traditional survey-based methods. The study contributes to the literature by providing a data-driven perspective on consumer experience in the Metaverse context and demonstrates the effectiveness of text mining techniques in analyzing large-scale consumer feedback. These findings also offer practical implications for firms seeking to enhance user experience and optimize product design in immersive digital environments.

Keywords: Immersive Brand Experience; Virtual Reality (VR); User-Generated Content (UGC); Text Mining; Sentiment Analysis; Natural Language Processing (NLP); Metaverse

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1 Introduction

Nowadays, with the rapid advancement of virtual reality (VR) and augmented reality (AR) technologies, immersive digital environments have become a key driver of innovation in marketing and consumer engagement. In recent years, the emergence of the Metaverse has further accelerated the integration of immersive technologies with brand communication, enabling consumers to interact with brands in highly interactive and multisensory virtual spaces. Unlike traditional one-way communication models, immersive environments allow users to actively participate in brand-related activities, thereby fostering deeper emotional connections and reshaping consumer – brand relationships. This transformation reflects a broader shift from transactional marketing to experiential and engagement-driven paradigms.

As immersive technologies continue to evolve, brands increasingly rely on virtual platforms to deliver experiential value through interactive content, virtual product trials, and personalized engagement. These developments not only enhance user participation and enjoyment but also generate large volumes of user-generated content (UGC), such as online reviews, comments, and feedback. Compared with structured survey data, UGC reflects consumers' authentic experiences and spontaneous emotional responses in real usage contexts, providing a valuable and dynamic data source for understanding immersive brand experiences. Moreover, the continuous accumulation of such data enables longitudinal and large-scale analysis of consumer perceptions in virtual environments.

Existing studies on brand experience and consumer behavior have predominantly relied on questionnaire-based approaches or experimental designs, which are often limited in capturing the multidimensional and dynamic nature of user experiences in virtual environments. In particular, research on immersive brand experience in the Metaverse context remains insufficient in leveraging unstructured data to explore consumers' perceptions and emotional evaluations. While prior studies have examined the roles of immersion, interactivity, and presence in shaping consumer behavior, there is still a lack of empirical research that systematically analyzes real-world consumer feedback using large-scale textual data.

Meanwhile, advances in natural language processing (NLP) and text mining techniques offer new opportunities to systematically analyze large-scale UGC and extract meaningful insights into consumer experience and sentiment. Techniques such as keyword extraction, sentiment classification, and semantic analysis enable researchers to uncover latent patterns in consumer feedback and provide a more comprehensive understanding of user perceptions. These methods are particularly suitable for analyzing emerging digital environments, where user interactions are complex, continuous, and data-intensive.

Against this backdrop, this study aims to explore the semantic characteristics and emotional tendencies of immersive brand experience by analyzing user reviews of mainstream VR devices. Specifically, this study collects online reviews of Meta Quest, HTC Vive, and PICO from Amazon and JD.com, and applies text mining techniques—including word segmentation, keyword extraction, and sentiment analysis—to identify key experiential dimensions and user concerns. By doing so, this research provides a data-driven understanding of how consumers perceive and evaluate immersive brand experiences in virtual environments.

This study contributes to the literature in two main ways. First, it extends existing research on immersive brand experience by incorporating UGC as a primary data source, thereby capturing more authentic and context-rich consumer insights and addressing the limitations of traditional survey-based approaches. Second, it demonstrates the applicability of text mining methods in brand research, offering a methodological framework for analyzing consumer experience in emerging digital environments such as the Metaverse. In addition, the findings provide practical implications for firms seeking to optimize immersive experiences and enhance consumer engagement through virtual platforms.

2 Theoretical and literature foundation

2.1 Immersive Brand Experience in Virtual Environments

With the advancement of virtual reality (VR), augmented reality (AR), and Metaverse technologies, consumer – brand interactions have evolved from passive information reception to active and multisensory engagement. Immersive brand experience refers to a technology-enabled experiential state in which consumers are deeply engaged in a virtual environment

and interact with brand-related stimuli through sensory, emotional, and cognitive channels. Rooted in flow theory and the experience economy, immersive experiences emphasize users' sense of presence, engagement, and emotional involvement in digitally constructed environments.

In virtual environments, immersion is often characterized by realism, interactivity, and perceptual intensity. These elements enable consumers to develop stronger emotional connections with brands and enhance their experiential evaluation. Compared with traditional media contexts, immersive brand experiences in the Metaverse provide more dynamic and personalized interaction scenarios, allowing users to participate in brand-related activities such as virtual exploration, product trials, and content co-creation. As a result, immersive environments play a critical role in shaping consumer perceptions and behavioral responses in digital marketing contexts.

2.2 User-Generated Content and Consumer Experience

User-generated content (UGC), including online reviews, comments, and shared experiences, has become a central source of information in digital consumption environments. Unlike firm-generated content, UGC reflects consumers' authentic perceptions, emotional responses, and usage experiences. In the context of immersive technologies, UGC provides valuable insights into how users perceive virtual interactions, evaluate product performance, and form experiential judgments.

Previous studies have shown that UGC plays a significant role in shaping brand perception and reducing information asymmetry. Online reviews, in particular, function as electronic word-of-mouth (eWOM), influencing consumers' decision-making processes and trust formation. Moreover, UGC often contains rich emotional and experiential information that cannot be fully captured through structured survey data. Therefore, analyzing UGC enables researchers to access real-time, context-specific consumer feedback and uncover multidimensional aspects of brand experience.

2.3 Text Mining and Sentiment Analysis in Brand Research

With the rapid growth of unstructured textual data, text mining has emerged as an important methodological approach in consumer and brand research. By applying natural language processing (NLP) techniques, researchers can systematically analyze large-scale textual data to extract key themes, semantic structures, and emotional tendencies. Common techniques include word segmentation, keyword extraction, sentiment analysis, and topic modeling.

In the context of brand research, text mining allows for the identification of consumers' core concerns, product evaluation dimensions, and affective responses. Sentiment analysis, in particular, enables the classification of emotional polarity in user reviews, providing insights into consumers' satisfaction, dissatisfaction, and underlying attitudes toward products and brands. Compared with traditional quantitative methods, text mining offers higher scalability and ecological validity, as it is based on naturally occurring data rather than controlled survey responses.

Furthermore, the integration of multiple text analysis techniques—such as combining keyword extraction with sentiment analysis—enhances the depth and reliability of findings. This methodological approach is particularly suitable for studying emerging digital environments like the Metaverse, where user interactions generate vast amounts of experiential data. As such, text mining provides a powerful tool for exploring the semantic and emotional dimensions of immersive brand experiences.

2.4 Research Gap

Despite the growing body of research on immersive technologies and brand experience, several gaps remain. First, existing studies have largely focused on structured data, such as surveys and experiments, while neglecting the potential of unstructured data sources like user reviews. Second, limited attention has been given to the semantic structure and emotional mechanisms underlying immersive brand experiences in virtual environments. Third, although text mining techniques have been increasingly applied in marketing research, their integration into studies of immersive brand experience remains underdeveloped.

To address these gaps, this study adopts a UGC-based text mining approach to analyze user reviews of VR devices and uncover the key experiential dimensions and emotional characteristics of immersive brand experience. By doing so, it contributes to both the theoretical understanding of consumer experience in virtual environments and the methodological advancement of data-driven brand research.

3 Research method

3.1 Data Collection

The data for this study were collected from user reviews of VR devices on two major e-commerce platforms, Amazon and JD.com. These platforms provide large volumes of user-generated content (UGC), including textual feedback and user ratings, which reflect consumers' real usage experiences in natural settings. Compared with structured survey data, such reviews capture spontaneous and context-rich user perceptions.

Specifically, reviews of three mainstream VR devices—Meta Quest, HTC Vive, and PICO—were collected using Octoparse, a web scraping tool designed for extracting large-scale online data. A total of XXXX user reviews were obtained between XXXX and XXXX, forming the dataset for subsequent analysis. Duplicate entries, irrelevant content, and incomplete records were removed to ensure data quality and reliability.

3.2 Text Analysis

This study applies natural language processing (NLP) techniques to analyze the review data and extract meaningful insights from unstructured text. First, Chinese word segmentation was conducted using the jieba library to preprocess the text and identify meaningful lexical units. To improve segmentation accuracy, domain-specific high-frequency terms related to VR devices (e.g., “tracking,” “immersion,” and “controller”) were incorporated into a custom dictionary.

Second, keyword extraction was performed using the term frequency – inverse document frequency (TF – IDF) method to identify representative and high-frequency terms that reflect users' primary concerns and experiential evaluations. This method allows for the identification of salient features within large-scale textual data.

Third, sentiment analysis was employed to classify the emotional polarity of reviews into positive, neutral, and negative categories. A lexicon-based approach was adopted to capture users' affective responses toward VR products and immersive experiences, enabling an overall assessment of user satisfaction and dissatisfaction.

3.3 Visualization

To facilitate interpretation and enhance the readability of results, word cloud visualization was employed to present high-frequency keywords and highlight dominant themes in user reviews. This visualization technique provides an intuitive representation of the semantic structure of the data and allows for rapid identification of key experiential elements.

3.4 Analytical Procedure

The analytical process consists of four main steps: data collection, text preprocessing, keyword extraction, and sentiment analysis. First, raw review data were collected and cleaned to remove noise and ensure consistency. Second, text preprocessing was conducted to segment words and construct a structured textual dataset. Third, keyword extraction was performed to identify core themes and experiential dimensions. Finally, sentiment analysis was conducted to examine emotional tendencies and user attitudes.

This systematic framework enables a comprehensive exploration of the semantic features and emotional tendencies of immersive brand experience based on UGC data, providing a solid methodological foundation for subsequent analysis.

4 Data analysis and results

4.1 Keyword Analysis

Based on the processed user review data, keyword extraction was conducted using the TF – IDF method to identify high-frequency and representative terms. The results indicate that users' evaluations of VR devices are primarily concentrated on several key aspects, including immersion, interaction, performance, and comfort. These dimensions reflect the core components of immersive brand experience in virtual environments and provide an empirical basis for understanding user perceptions.

Table 1 presents the high-frequency keywords extracted from user reviews of the three VR devices. As shown in Table 1, terms related to immersive experience (e.g., “immersion,” “experience,” and “visual”) appear most frequently,

indicating that users place significant importance on the sense of presence and realism in virtual environments. This finding suggests that immersive perception is a central factor influencing user satisfaction and experiential evaluation.

In addition, interaction-related keywords (e.g., “control,” “tracking,” and “operation”) highlight the importance of usability and system responsiveness in shaping user experience. Efficient and intuitive interaction mechanisms enable users to engage more effectively with virtual environments, thereby enhancing overall engagement and perceived value.

Performance-related terms such as “resolution,” “battery,” and “latency” also emerge as major concerns, reflecting users’ strong attention to technical quality and device functionality. These results indicate that, despite the importance of immersive experience, functional performance remains a fundamental prerequisite for positive user evaluation.

Furthermore, comfort-related keywords (e.g., “comfort,” “weight,” and “fatigue”) reveal that physical usability plays a crucial role in prolonged usage scenarios. This suggests that immersive brand experience is not only determined by virtual interaction but also influenced by ergonomic and hardware-related factors.

Overall, the keyword analysis demonstrates that immersive brand experience in VR contexts is a multidimensional construct shaped by both experiential and functional attributes, highlighting the need to balance immersive features with technical performance and user comfort.

Table 1. High-Frequency Keywords of VR User Reviews Across Devices

Category	Representative Keywords	Interpretation
Immersion	immersion, experience, visual, realistic	Users emphasize the sense of presence and realism in virtual environments
Interaction	control, tracking, operation, response	Interaction quality and system responsiveness are key concerns
Performance	resolution, battery, latency, graphics	Technical performance significantly affects user experience
Comfort	comfort, weight, wearing, fatigue	Physical comfort influences prolonged usage experience
Content/Usage	games, applications, entertainment	Users frequently mention usage scenarios and content availability

To further illustrate the distribution of keywords, word cloud visualization was conducted. As shown in Figure 1, the most prominent words are related to immersion, interaction, and performance, confirming the results of keyword analysis.



Figure 1. Word Cloud of VR User Reviews Across Devices

4.2 Sentiment Analysis

Sentiment analysis was conducted to examine the emotional tendencies of user reviews. The results indicate that the majority of reviews exhibit positive sentiment, suggesting a generally high level of user satisfaction with VR devices and immersive experiences.

Positive evaluations are mainly associated with strong immersion, enjoyable interaction, and innovative features. In contrast, negative sentiments are primarily related to technical limitations, including device performance issues, discomfort during prolonged use, and usability challenges.

These findings suggest that while immersive experiences enhance overall user satisfaction, functional deficiencies may still negatively affect user perceptions.

4.3 Comparative Analysis Across Devices

A comparative analysis of the three VR devices—Meta Quest, HTC Vive, and PICO—reveals differences in user evaluations across brands. While all devices receive positive feedback regarding immersive experience, variations exist in terms of performance and usability.

Table 2 summarizes the key experiential dimensions identified for each device. As shown in Table 2, some devices are more strongly associated with high immersion and visual quality, whereas others are more closely linked to comfort and accessibility. These differences reflect the influence of technological characteristics on user experience and highlight the diversity of consumer perceptions across brands.

Table 2. Comparison of Key Experience Dimensions Across VR Devices

Dimension	Meta Quest	HTC Vive	PICO
Immersion	Mixed-reality immersion; strong visual experience and spatial reconstruction	High immersion with room-scale tracking and large interaction space	Moderate immersion with emphasis on audio-visual entertainment
Interaction	Smooth interaction; controller tracking and passthrough features	High-precision tracking; complex interaction system	Simple and user-friendly interaction; convenient operation
Performance	High resolution and system stability; occasional battery limitations	Professional-grade performance; high hardware requirements	Balanced performance; optimized for lightweight use
Comfort	Moderate comfort; influenced by device weight and battery usage	Lower comfort due to complex setup and equipment constraints	High comfort; lightweight design suitable for long-term use
Usability	Easy to use; relatively simple setup	Complex installation and configuration	Highly accessible; ready-to-use experience
Main User Concerns	Battery life, performance fluctuations	Installation complexity, compatibility issues	Comfort, content variety, and usability
Key Experience Focus	Balanced performance and immersive interaction	High-end immersive and professional experience	Comfort-oriented and user-friendly experience

4.4 Summary of Findings

Overall, the analysis reveals that immersive brand experience in VR contexts is characterized by four core dimensions: immersion, interactivity, performance, and comfort. These dimensions jointly shape users' perceptions and evaluations of VR devices, highlighting the multidimensional nature of immersive experience in virtual environments. Among them, immersion and interactivity play a dominant role in enhancing user engagement, while performance and comfort serve as essential supporting factors that influence overall satisfaction.

User evaluations are predominantly positive, indicating a generally favorable perception of VR devices and immersive experiences. However, these evaluations are also significantly influenced by functional and technical factors, such as device performance, usability, and physical comfort. This suggests that while immersive technologies enhance experiential value, their effectiveness is contingent upon reliable technical performance and user-centered design.

Furthermore, the combined use of keyword analysis, sentiment analysis, and visualization provides a comprehensive understanding of the semantic structure and emotional characteristics of user-generated content. This integrated analytical approach enables the identification of both experiential dimensions and underlying emotional tendencies, offering a more holistic perspective on consumer experience.

These findings provide empirical evidence for understanding immersive brand experience from a data-driven perspective and contribute to the growing body of research that leverages unstructured data in marketing analysis. Moreover, they lay a solid foundation for further investigation into the relationships between immersive experience, brand trust, and brand loyalty in virtual environments, particularly in the context of the Metaverse.

5 Conclusion

This study explores the characteristics of immersive brand experience in virtual environments by analyzing user-generated content (UGC) from VR device reviews. Based on textual data collected from Amazon and JD.com, natural language processing (NLP) techniques were applied to identify key experiential dimensions and emotional tendencies associated with three mainstream VR devices: Meta Quest, HTC Vive, and PICO.

The results reveal that immersive brand experience is primarily characterized by four core dimensions: immersion, interaction, performance, and comfort. Among these, immersion and interaction emerge as the most prominent factors shaping user evaluations, highlighting the importance of presence and real-time responsiveness in virtual environments. At the same time, performance and comfort play a critical role in influencing overall user satisfaction, particularly in terms of device functionality and long-term usability.

In addition, sentiment analysis indicates that user evaluations are predominantly positive, suggesting a generally high level of satisfaction with VR devices and immersive experiences. However, negative feedback is mainly associated with technical limitations, including performance issues, usability challenges, and discomfort during prolonged use. These findings demonstrate that while immersive technologies enhance user engagement and experiential value, functional deficiencies remain an important constraint on user perception and acceptance.

Furthermore, comparative analysis across devices reveals that user perceptions vary depending on technological characteristics. Meta Quest is associated with balanced performance and interactivity, HTC Vive with high immersion and professional-level precision, and PICO with comfort and accessibility. This highlights the heterogeneous nature of user experience across different VR platforms and suggests that different technological strategies lead to distinct experiential outcomes.

From a theoretical perspective, this study contributes to the literature by providing a data-driven understanding of immersive brand experience based on UGC, extending existing research beyond traditional survey-based approaches. It enriches the conceptualization of immersive experience by identifying its key dimensions from naturally occurring consumer data. Methodologically, this study demonstrates the applicability and effectiveness of text mining techniques in analyzing large-scale consumer feedback in emerging digital environments such as the Metaverse.

From a managerial perspective, the findings suggest that firms should not only focus on enhancing immersive features but also pay close attention to usability and comfort to improve overall user satisfaction. Balancing technological advancement with user-centered design is essential for strengthening brand perception and sustaining consumer engagement in virtual environments.

Despite these contributions, this study has several limitations. The analysis is based on review data from selected platforms and may not fully capture all user groups or usage contexts. In addition, the study focuses on textual data and does not incorporate multimodal information such as images or behavioral data. Future research may integrate multiple data sources, including behavioral tracking and visual analytics, and apply advanced analytical methods such as deep learning to further explore consumer experience in immersive environments.

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